

14.H1 Creating Relative Frequency Tables

Lesson Introduction

 Benchmarks	MA.912.DP.3.2 Given marginal and conditional relative frequencies, construct a two-way relative frequency table summarizing categorical bivariate data.			 Learning Target	I can construct a two-way relative frequency table.
	MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate.				
	MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, and whether it is univariate or bivariate, and interpret the different components and quantities in the display.				
 Benchmark Coherence	Building On	Working Toward			
	N/A	MA.912.DP.4.3 Calculate the conditional probability of two events and interpret the result in terms of its context.			
	MA.912.DP.4.4 Interpret the independence of two events using conditional probability.				
 Essential Question	How do you construct a two-way relative frequency table?				
 Lesson Narrative	In this lesson, students learn about joint, marginal, and conditional relative frequencies and how to construct two-way relative frequency tables. During the lesson, students will also learn how to identify conditional relative frequencies and how to use them to find missing joint relative frequency values.				
 Component Summary	14.H1.1 Warm-Up (~5 min.)	Percentage activity (SE p. 325)	14.H1.4 Guided Practice (10-15 min.)	Practice using data to create a tree diagram and a two-way frequency table (SE p. 329)	
	14.H1.2 Exploration (10-15 min.)	Exploring joint and conditional relative frequencies (SE p. 326)	14.H1.5 Try It (10 min.)	Students independently create a two-way frequency table (SE p. 330)	
	14.H1.3 Guided Instruction (10-15 min.)	Introduction to using data to create a two-way frequency table (SE p. 327)	14.H1.6 Wrap-Up (~5 min.)	Growth mindset self-reflection (SE p. 330)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Joint Relative Frequency - Marginal Relative Frequency - Conditional Relative Frequency <u>Additional Terms:</u> N/A		N/A		(Optional) Anchor Chart for joint, marginal, and conditional relative frequencies

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may use conditional relative frequencies to fill in a two-way relative frequency table instead of using those to find the missing joint relative frequency values.	- Consider working out 14.H1.5 Try It before the lesson so that you are aware of the challenges students may face when trying to create the two-way relative frequency table. - This lesson is intended for students enrolled in Algebra 1 Honors, but can be used as enrichment for students enrolled in Algebra 1.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Collect and Display During this lesson, students will practice their ability to read a context to collect the needed data to display it in a two-way relative frequency table. The lesson provides that support needed during 14.H1.3 Guided Instruction for students to be aware of the language in the context to determine when provided a joint, marginal, or conditional relative frequency.	MA.K12.MTR.2.1: Participate in Effortful Learning During 14.H.1.5 Try It, students will build perseverance by modifying methods as needed, while solving a challenging task. During this task, students will not be provided data that will not directly fit into the two-way relative frequency table. They will have to develop their ability to analyze the data provided in order to complete the table.
	Consider creating, or having students create, an Anchor Chart for joint, marginal, and conditional relative frequencies. On the Anchor Chart, provide examples of each so that students can refer to them during 14.H1.4 Guided Practice and 14.H1.5 Try It portions of the lesson.	
Lesson Notes:		

14.H1.1 Warm-Up: What Percentage?

Component Overview: Students calculate marginal and joint relative frequencies in 14.H1.1 Warm-Up.



14.H1.1 Warm-Up: What Percentage?

- Mindy is in charge of ordering supplies for the break room at her workplace, so she surveys her coworkers. She asks them if they prefer coffee or tea and whether or not they like sugar in their beverages. Her findings are shown in the two-way table.

	Coffee	Tea	Total
Sugar	15	8	23
No Sugar	25	2	27
Total	40	10	50

Use the two-way table to answer the following questions.

- What percentage of people surveyed prefer coffee?

The percentage of people surveyed who prefer coffee is $\frac{40}{50} = \frac{80}{100} = 80\%$.

- What percentage of people surveyed drink tea with sugar?

The percentage of people surveyed who drink tea with sugar is $\frac{8}{50} = \frac{16}{100} = 16\%$.



Use informal observation of this activity of those who are struggling versus proficient to inform small groups during the independent practice. Students who struggle may need additional support during 14.H1.5 Try It portion of this lesson.

14.H1.2 Exploration: Two-Way Relative Frequency Tables

Component Overview: In 14.H1.2 Exploration, students learn about two-way relative frequency tables and how the frequency table can be used to find the conditional relative frequency from a given context.



14.H1.2 Exploration: Two-Way Relative Frequency Tables

A 2018 Pew Research Center survey of American teenagers was conducted to analyze trends in social media and technology. 741 teens who use social media were asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral) and how much time they believe they spend with their friends in person (too much, too little, or about the right amount). The results are summarized in the joint frequency table.



Consider facilitating a conversation by prompting students to read and analyze the information on this page. Without giving away any answers to questions 1-5, ask students about what they notice and wonder about the 2 two-way relative frequency tables.

		Effect on People Your Age			
Time spent With Friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

The data can also be displayed in a joint relative frequency table, where the cells represent the joint and marginal relative frequencies. These are often useful to construct when given percentages for a data set, rather than frequencies.

		Effect on People Your Age			
Time spent With Friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	2.97%	2.83%	2.7%	8.5%
	Too Little	11.74%	10.39%	14.17%	36.3%
	Just Right	14.84%	15.25%	25.10%	55.20%
	Total	29.55%	28.48%	41.97%	100.00%

14.H1.2 Exploration: Two-Way Relative Frequency Tables (continued)



A **joint relative frequency** is the ratio of a joint frequency and the total number of data points.

A **marginal relative frequency** is the ratio of a marginal frequency and the total number of data points.

1. What is the sum of the joint relative frequencies?
The sum of all of the joint relative frequencies is 100%.
2. What is the sum of the marginal relative frequencies for the variable “effect on people your age?”
The sum of the marginal relative frequencies for the variable “effect on people your age” is 100%.
3. What is the sum of the joint relative frequencies for those who believe the effect of social media is mostly positive?
The sum of the joint relative frequencies for those who believe the effect of social media is mostly positive is 29.55%.
4. What is the sum of the joint relative frequencies for those who believe they spend too much time with friends in person?
The sum of the joint relative frequencies for those who believe they spend too much time with friends in person is 8.5%.



After reading and analyzing the information on the previous page, consider releasing students to work on questions 1-4, then pause students before they get to question 5. For question 5, point out to students why the percentages in the joint relative frequency table are not the percentages they are looking for here. Students may try to state that the percentage is 2.97%, instead of dividing 22 by 63 to find the correct percentage of 34.92%.

14.H1.2 Exploration: Two-Way Relative Frequency Tables (continued)

5. Use only the values from the joint frequency table to answer the following questions.
- What percentage of those who spend too much time with friends in person believe that the effect of social media has been mostly positive?
Of the teens surveyed who say they spend too much time with friends in person, 34.92% of them believe the effect of social media on people their age has been mostly positive.
 - What percentage of those who spend too much time with friends in person believe that the effect of social media has been mostly negative?
Of the teens surveyed who spend too much time with friends in person, 33.33% believe that the effect of social media on people their age has been mostly negative.

The percentages calculated in Parts A and B above are **conditional relative frequencies**.



A **conditional relative frequency** is the ratio of a joint relative frequency to a marginal relative frequency. Equivalently, the ratio of a joint frequency and a marginal frequency.



Ask students to relate the types of relative frequency to their names – joint relative frequencies show where the two variables join, marginal relates to the margins of the tables, and conditionals relate to situations where there is a condition placed on the statement. For example, in question 5a, we calculated the percentage of people who believe the effect of social media is mostly positive **of those who spend too much time with friends**. The condition is people who spend too much time with friends, so that is what is used as the total in those cases.

- What is the condition in question 5b?

14.H1.3 Guided Instruction: Creating a Two-Way Relative Frequency Table

Component Overview: In 14.H1.3 Guided Instruction, students will learn how to use conditional relative frequencies to find missing joint relative frequency values.



14.H1.3 Guided Instruction: Creating a Two-Way Relative Frequency Table

Consider the following data on whether or not animals received a microchip from the Kirkman Road Veterinary Clinic's database of canines (dogs) and felines (cats).

- 37.61% of animals in the sample were felines.
- 30.63% of animals in the sample did not have microchips.
- Of the animals with microchips, 81.17% were canines.
- 34.73% of the felines had microchips.

1. Complete each statement.

a. 37.61% represents a frequency.

- ☐ joint
☒ marginal
☐ conditional

relative

b. 30.63% represents a frequency.

- ☐ joint
☒ marginal
☐ conditional

relative

c. 81.17% represents a frequency.

- ☐ joint
☐ marginal
☒ conditional

relative

d. 34.73% represents a frequency.

- ☐ joint
☐ marginal
☒ conditional

relative

2. Explain how to determine the relative frequency of microchipped canines in the sample.

To determine the relative frequency of microchipped canines in the sample, multiply the percentage of microchipped animals from the study, 69.37%, and percentage of canines who are microchipped, 81.17%.



Consider having students start to complete the two-way relative frequency in question 4, using only the marginal relative frequency data as part of question 2. This will help students visualize the missing data, how to interpret the solution to question 2 by finding 81.17% of 69.37%, and why it would be a joint relative frequency.

14.H1.3 Guided Instruction: Creating a Two-Way Relative Frequency Table (continued)

3. The value in question 2 represents a relative frequency.

- ☐ joint
- ☐ marginal
- ☐ conditional

4. Use the data to create a joint relative frequency table. Space is provided to make any calculations.

$$37.61\% \cdot 34.73\% = 13.06\%$$

$$37.61\% - 13.06\% = 24.55\%$$

$$69.37\% \cdot 81.17\% = 56.31\%$$

Clients at Kirkman Road Veterinary Clinic

	Microchipped	Not Microchipped	Total
Feline	13.06%	24.55%	37.61%
Canine	56.31%	6.08%	62.39%
Total	69.37%	30.63%	100%

5. Complete the statement.
The smallest percentage of animals in the sample were

- ☐ microchipped
- ☐ not microchipped

- ☐ felines.
- ☐ canines.

6. What percentage of canines had microchips?
The percentage of canines that had microchips is 90.25%. This can be calculated
7. Which kind of pet is more likely to be microchipped?
Explain your reasoning.
According to this data, canines are more likely to be microchipped. Only 34.73% of felines in the study were microchipped, and a much larger percentage, 90.25%, of canines in the study were microchipped.



Students may take a different approach than the one provided to find all the missing percentages. Ask students to share their method for finding the missing values.



To encourage students to check their solutions, consider asking the following questions:

- Did your marginal relative frequencies add up to 100%?
- Did your joint relative frequencies add up to 100%?
- Where do you think a mistake could have happened?



For a challenge, have students write their own questions based on the complete two-way relative frequency table for their partners to answer.

14.H1.4 Guided Practice: Creating a Two-Way Relative Frequency Table

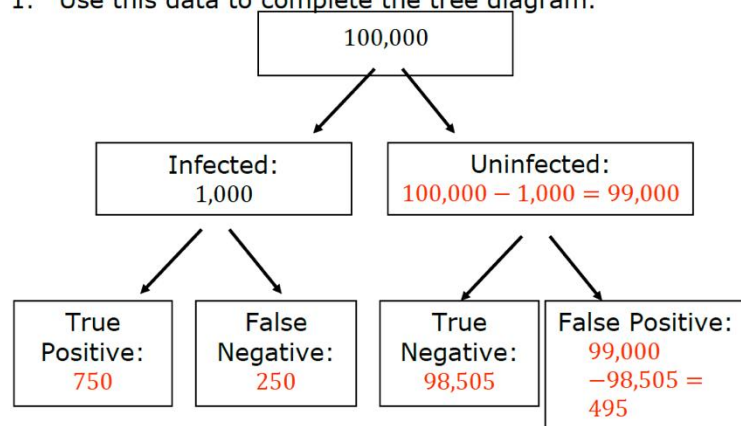
Component Overview: Students will use the data provided from the context to complete a tree diagram that will then be used to help students to create a two-way relative frequency table.



14.H1.4 Guided Practice: Creating a Two-Way Relative Frequency Table

A study is performed to determine the efficacy of a new test that determines whether or not a patient has a disease. 100,000 people were included in the study. 1% of the participants are infected with the disease. Of those infected, the test correctly identified the presence of the disease in 750 patients, and 250 patients received a false negative. For those patients that were not infected with the disease, the test was correct in 98,505 cases.

- Use this data to complete the tree diagram.



- Complete the two-way relative frequency table to represent the data.

	Positive	Negative	Total
Infected	0.75%	0.25%	1%
Not Infected	0.495%	98.505%	99%
Total	1.245%	98.755%	100%



Discuss what efficacy means – the ability to produce a desired or intended result.

- What do you think it means for a study to determine the efficacy of a new test?



Students may mistakenly divide by 1,000 or 99,000 to find the joint relative frequencies. Consider asking the following questions:

- When creating two-way relative frequency tables what is the data relative to?
- How is that different from a conditional relative frequency?

14.H1.5 Try It: RMS Titanic

Component Overview: Students use data provided from a context to complete a two-way frequency table. In this activity, students will have to use the provided data to find missing pieces of the two-way relative frequency table.



14.H1.5 Try It: RMS Titanic

- The RMS Titanic was a British passenger liner operated by the White Star Line that sank in the North Atlantic Ocean on April 15, 1912, after striking an iceberg during her maiden voyage from Southampton to New York City. Of the estimated 2,201 passengers and crew aboard, 1,490 died, making the sinking, at the time, one of the deadliest of a single ship.

Passengers can be classified as whether the person survived (Yes or No) and the class of the ticket they had (1st, 2nd, 3rd, or Crew).

- 14.8% of passengers had 1st class tickets.
- 13.0% of passengers had 2nd class tickets.
- 35.4% of those who died were third class passengers.
- 25.0% of those who survived were 3rd class passengers.
- 76.0% of the crew died.
- 62.5% of 1st class passengers survived.

Create a two-way relative frequency table based on the data provided.



Consider using informal observations from 14.H1.1 Warm-Up as well as student performance during the 14.H1.3 Guided Instruction to pull a small group of students for additional support.



Students may only look at the bulleted information and problem context. Refer students back to the context to find other marginal relative frequencies as they create the two-way frequency table.

Ticket	Survived		Total
	No	Yes	
1st	5.55%	9.25%	14.8%
2nd	7.67%	5.33%	13.0%
3rd	23.97%	8.08%	32.05%
Crew	30.51%	9.64%	40.15%
Total	67.7%	32.3%	100.0%

14.H1.6 Wrap-Up: Reflection

Component Overview: Students complete a growth mindset self-reflection about today's lesson.



14.H1.6 Wrap-Up: Reflection

1. Complete each statement.
 - a. After today's lesson, I am still unsure about ...
Answers will vary.
 - b. To fill this gap, I intend to ...
Answers will vary.



Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.



Consider our online resources for additional remediation opportunities for students to practice or test their abilities.